H Synthetic Supervision Details

We can use $\text{PRM}_{\text{large}}$ to provide either outcome or process supervision for smaller models. We determine the labels for individual steps based on the step-level probabilities outputted by $\text{PRM}_{\text{large}}$. To do this, we set an arbitrary threshold: any step that $\text{PRM}_{\text{large}}$ assigns a negative label with greater than 20% probability is considered incorrect. We choose this threshold based on the observation that $\text{PRM}_{\text{large}}$ is slightly miscalibrated in the direction of favoring positive labels.

To provide process supervision for a solution, we directly return the step-level labels (positive or negative) provided by $\text{PRM}_{\text{large}}$, up until the first step that is marked as negative. This mimics our true human data collection process. To provide outcome supervision, we mark the solution as correct if and only if $\text{PRM}_{\text{large}}$ considers every step to be correct (using the same thresholding logic).

I PRM Visualizations

All examples shown come from the large-scale generator (GPT-4). We note the pass-rate under the generator to give some sense of the difficulty of these problems.

I.1 True Positives

These cherry-picked examples show the best-of-1860 solution from the generator as ranked by the large-scale PRM.

Problem 1. Generator pass-rate: 0.1%. This challenging trigonometry problem requires applying several identities in a not-at-all obvious succession. Most solution attempts fail, because it is hard to choose which identities are actually helpful. Though successful solutions to this problem are rare, the reward model correctly recognizes when a valid chain-of-thought has been found.

Simplify $\tan 100^\circ + 4\sin 100^\circ$.

I notice that the angle 100 degrees is not a special angle, so I will need to use some trigonometric identities to simplify this expression.

I recall that $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\sin 2\theta = 2\sin\theta\cos\theta$.

I can use these identities to rewrite the expression as $\frac{\sin 100^\circ}{\cos 100^\circ} + 4\sin 100^\circ$.

I can combine the terms by finding a common denominator of $\cos 100^\circ$.

I get $\frac{\sin 100^\circ + 4\sin 100^\circ \cos 100^\circ}{\cos 100^\circ}$.

I can simplify the numerator by using the double angle identity again, but in reverse.

I get $\frac{\sin 100^\circ + 2\sin 200^\circ}{\cos 100^\circ}$.

I notice that 200 degrees is the same as 180 degrees plus 20 degrees, so I can use the sum angle identity for sine.

I get $\frac{\sin 100^\circ + 2(\sin 180^\circ \cos 20^\circ + \cos 180^\circ \sin 20^\circ)}{\cos 100^\circ}$.

I recall that $\sin 180^\circ = 0$ and $\cos 180^\circ = -1$, so I can simplify further.

I get $\frac{\sin 100^\circ - 2\sin 20^\circ}{\cos 100^\circ}$.

I notice that 100 degrees is the same as 90 degrees plus 10 degrees, so I can use the sum angle identity for sine and cosine.

I get $\frac{(\sin 90^\circ \cos 10^\circ + \cos 90^\circ \sin 10^\circ) - 2\sin 20^\circ}{(\cos 90^\circ \cos 10^\circ - \sin 90^\circ \sin 10^\circ)}$.

I recall that $\sin 90^\circ = 1$ and $\cos 90^\circ = 0$, so I can simplify further.

I get $\frac{\cos 10^\circ - 2\sin 20^\circ}{-\sin 10^\circ}$.

I can simplify the sign by multiplying the numerator and denominator by -1.

I get $\frac{2\sin 20^\circ - \cos 10^\circ}{\sin 10^\circ}$.

I notice that 20 degrees is the same as 30 degrees minus 10 degrees, so I can use the difference angle identity for sine.

I get $\frac{2(\sin 30^\circ \cos 10^\circ - \cos 30^\circ \sin 10^\circ) - \cos 10^\circ}{\sin 10^\circ}$.

I recall that $\sin 30^\circ = \frac{1}{2}$ and $\cos 30^\circ = \frac{\sqrt{3}}{2}$, so I can simplify further.

I get $\frac{\cos 10^\circ - \sqrt{3}\sin 10^\circ - \cos 10^\circ}{\sin 10^\circ}$.

I can cancel out the $\cos 10^\circ$ terms in the numerator.

I get $\frac{-\sqrt{3}\sin 10^\circ}{\sin 10^\circ}$.

I can cancel out the $\sin 10^\circ$ terms in the numerator and denominator.

I get $-\sqrt{3}$.

Answer: $-\sqrt{3}$

Problem 2. Generator pass-rate: 5.8%. In step 7 and 8, the generator starts performing guess-and-check. This is a common place the model might hallucinate, by claiming a particular guess is successful when it isn't. In this case, the reward model verifies each step and determines that the chain-of-thought is correct.

Find the unique odd integer t such that $0 < t < 23$ and $t + 2$ is the inverse of t modulo 23.

I need to find an odd integer t that satisfies the congruence $t + 2 \equiv t^{-1} \pmod{23}$.

To do that, I can multiply both sides by t and get $t^2 + 2t \equiv 1 \pmod{23}$.

Then I can complete the square by adding and subtracting $(2/2)^2 = 1$ to the left side and get $(t + 1)^2 - 1 \equiv 1 \pmod{23}$.

Simplifying, I get $(t + 1)^2 \equiv 2 \pmod{23}$.

Now I need to find a square root of 2 modulo 23.

I can try some values of $t + 1$ and see if they work.

For example, if $t + 1 = 3$, then $(t + 1)^2 = 9 \not\equiv 2 \pmod{23}$.

If $t + 1 = 5$, then $(t + 1)^2 = 25 \equiv 2 \pmod{23}$.

So one possible value of t is $t = 5 - 1 = 4$.

But this is not odd, so I need to find another one.

I know that if $t + 1$ is a square root of 2 modulo 23, then so is $-t - 1$.

So I can try $-t - 1 = 5$ and get $t = -6$.

But this is negative, so I need to add 23 to get a positive value.

So another possible value of t is $t = -6 + 23 = 17$.

This is odd and satisfies the original congruence, so this is the unique solution.

Answer: 17

Problem 3. Generator pass-rate: 1.7%. The generator successfully applies several trigonometric identities to simplify the expression.

Find the degree measure of the least positive angle θ for which

$$\tan\theta = \frac{\cos 5^\circ \cos 20^\circ + \cos 35^\circ \cos 50^\circ - \sin 5^\circ \sin 20^\circ - \sin 35^\circ \sin 50^\circ}{\sin 5^\circ \cos 20^\circ - \sin 35^\circ \cos 50^\circ + \cos 5^\circ \sin 20^\circ - \cos 35^\circ \sin 50^\circ}.$$

I notice that the numerator and denominator of the fraction are both sums of products of cosine and sine functions of angles that add up to 25 degrees or 85 degrees.

This suggests that I might be able to use some trigonometric identities to simplify the expression.

For example, I recall that $\cos(A + B) = \cos A \cos B - \sin A \sin B$ and $\sin(A + B) = \sin A \cos B + \cos A \sin B$.

If I apply these identities to the terms in the numerator, I get $\cos 5^\circ \cos 20^\circ + \cos 35^\circ \cos 50^\circ - \sin 5^\circ \sin 20^\circ - \sin 35^\circ \sin 50^\circ$ $= \cos(5^\circ + 20^\circ) + \cos(35^\circ + 50^\circ) = \cos 25^\circ + \cos 85^\circ$.

Similarly, if I apply these identities to the terms in the denominator, I get $\sin 5^\circ \cos 20^\circ - \sin 35^\circ \cos 50^\circ + \cos 5^\circ \sin 20^\circ$ $- \cos 35^\circ \sin 50^\circ = \sin(5^\circ + 20^\circ) - \sin(35^\circ + 50^\circ) = \sin 25^\circ - \sin 85^\circ$.

Now I have $\tan\theta = \frac{\cos 25^\circ + \cos 85^\circ}{\sin 25^\circ - \sin 85^\circ}$.

I wonder if I can simplify this fraction further.

I recall another trigonometric identity that says $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$.

If I apply this identity to the numerator, I get $\cos 25^\circ + \cos 85^\circ = 2\cos\frac{25^\circ+85^\circ}{2}\cos\frac{25^\circ-85^\circ}{2} = 2\cos 55^\circ \cos(-30^\circ)$.

I also recall that $\cos(-\theta) = \cos\theta$, so I can write $\cos(-30^\circ) = \cos 30^\circ$.

Similarly, I recall another trigonometric identity that says $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$.

If I apply this identity to the denominator, I get $\sin 25^\circ - \sin 85^\circ = 2\cos\frac{25^\circ+85^\circ}{2}\sin\frac{25^\circ-85^\circ}{2} = 2\cos 55^\circ \sin(-30^\circ)$.

I also recall that $\sin(-\theta) = -\sin\theta$, so I can write $\sin(-30^\circ) = -\sin 30^\circ$.

Now I have $\tan\theta = \frac{2\cos 55^\circ \cos 30^\circ}{2\cos 55^\circ(-\sin 30^\circ)} = -\frac{\cos 30^\circ}{\sin 30^\circ}$.

I know that $\cos 30^\circ = \frac{\sqrt{3}}{2}$ and $\sin 30^\circ = \frac{1}{2}$, so I can simplify this to $\tan\theta = -\frac{\sqrt{3}}{1} = -\sqrt{3}$.

I also know that $\tan\theta$ is negative in the second and fourth quadrants, and that $\tan\theta = -\sqrt{3}$ when θ is 120° or 300°.

Therefore, the least positive angle θ for which $\tan\theta = -\sqrt{3}$ is 120°.

Answer: 120